Software Requirements Specification

for

E-Learning Management System

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Ashwita Pawar	03-08-2026	Initial draft	1.0
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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and nonfunctional requirements for Release 1.0 of the **E-Learning Management System (E-LMS)**. This document covers the core learning platform modules, including Course Creation, Assignment Submissions, Quizzes, Attendance Tracking, and Automated Grading. It serves as the official baseline for system design, implementation, testing, and project verification.

1.2 Document Conventions

Functional requirements are identified with a dotted, feature-based tag (e.g., Assign.Submit.Late) so that each requirement can be traced back to the system feature it belongs to. Nonfunctional requirements use a short prefix followed by a sequence number (e.g., PE-1 for a Performance requirement, SE-1 for a Security requirement). Unless a requirement explicitly states otherwise, all requirements in this document are of High priority and are committed for Release 1.0.

1.3 Product Scope

The E-LMS will allow an educational institution to manage course delivery entirely online. Instructors will be able to create and organize courses, publish learning material, create and grade assignments and quizzes, and track student attendance. Students will be able to enroll in courses, view material, submit assignments, attempt quizzes, participate in discussion forums, and receive notifications about deadlines and grades. Release 1.0 covers eight core feature areas: Course Creation and Management, Assignment Submission, Quizzes, Attendance Tracking, Automated Grading, Discussion Forums, Notifications, and Video Conferencing / Live Classes. Payment/enrollment fee processing and completion certificates are explicitly out of scope for Release 1.0.

1.4 References

1. IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.

2. Wiegers, Karl. Software Requirements Specification Template, Process Impact, www.processimpact.com.

2. Overall Description

2.1 Product Perspective

The E-LMS is a new, self-contained, web-based system. It is not a follow-on to an existing product and does not integrate with any external Student Information System (SIS), inventory system, or payment gateway in Release 1.0. All user accounts, course enrollments, and academic records are created and maintained within the E-LMS itself. Live video sessions are conducted through an external conferencing provider (exact provider TBD; see Appendix C); the E-LMS stores and displays the session link and schedule but does not host the video stream itself.

2.2 Product Functions

The major functions the E-LMS must provide are summarized below; details are provided in Section 4.

- Create, organize, and publish courses and course material (**Instructor**).
- Enroll in courses and access published material (**Student**).
- Create, submit, and grade assignments, including late-submission handling.
- Create timed, auto-graded quizzes and attempt them within configured limits.
- Mark and report on Student attendance per course session.
- Automatically compute grades and maintain a weighted per-course gradebook.
- Post and reply to discussion threads within a course forum.
- Send email notifications for deadlines, released grades, and announcements.
- Schedule and join live class sessions through an external video-conferencing link.

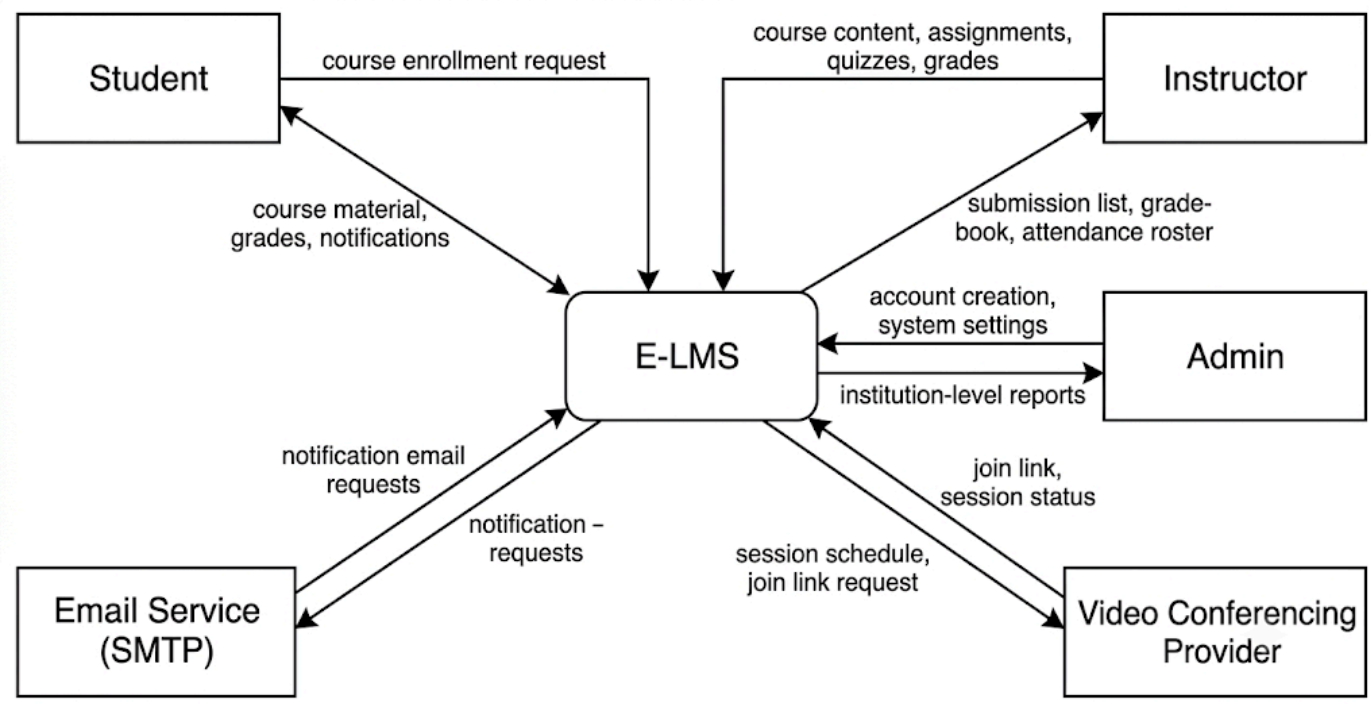


Figure 1. Context diagram for Release 1.0 of the E-LMS.

2.3 User Classes and Characteristics

Student — A learner who enrolls in one or more courses to access material, submit assignments, attempt quizzes, view grades and attendance, participate in discussion forums, and join live class sessions. Students are the most frequent users of the system.

Instructor — A faculty member who creates and manages courses, publishes course material, creates assignments and quizzes, grades submitted work, marks attendance, moderates discussion forums, and schedules live class sessions. An Instructor has full control over the courses they own.

Admin — A system administrator responsible for creating and managing user accounts (Students, Instructors), assigning Instructors to courses, configuring system-wide settings, and generating institution-level reports. Admins do not typically create course content or grade student work.

2.4 Operating Environment

OE-1: The E-LMS shall operate correctly on the current stable releases of Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.

OE-2: The E-LMS shall operate on a Linux-based application server running a current LTS release, with a relational database backend.

OE-3: The E-LMS shall be accessible over the public Internet via HTTPS, requiring no client side installation beyond a standard web browser.

OE-4: The E-LMS user interface shall be responsive and usable on desktop, tablet, and mobile-sized browser viewports.

2.5 Design and Implementation Constraints

CO-1: The system shall store all passwords using a salted, industry-standard one-way hashing algorithm; plaintext passwords shall never be stored.

CO-2: All file uploads (assignment submissions, course material) shall be scanned for allowed file types and shall be limited to a configurable maximum file size.

CO-3: The system's date and time values shall be stored in UTC and displayed to each user in their configured local time zone.

CO-4: The system shall be implemented using a relational database to preserve referential integrity between courses, enrollments, submissions, and grades.

2.6 User Documentation

UD-1: The system shall provide role-specific online help (Student, Instructor, Admin) accessible from every screen.

UD-2: The first time a new user logs in, the system shall offer a short guided walkthrough of the features available to that user's role.

2.7 Assumptions and Dependencies

AS-1: It is assumed that Students, Instructors each have reliable Internet access and a device with a modern web browser.

AS-2: It is assumed that the institution issues each user a unique institutional email address, which is used for account verification and notifications.

DE-1: The live-class feature depends on an external video-conferencing provider (TBD)

3. External Interface Requirements

3.1 User Interfaces

UI-1: The E-LMS screens shall follow a single, consistent visual style (typography, color, navigation placement) across all pages.

UI-2: Every screen shall provide a help link explaining the function of that screen.

UI-3: All Student-facing and Instructor-facing pages shall be fully operable using the keyboard alone, in addition to mouse input.

3.2 Hardware Interfaces

No hardware interfaces have been identified. The E-LMS is accessed entirely through standard web browsers on general-purpose computing devices.

3.3 Software Interfaces

SI-1: Email Service

SI-1.1: The system shall send notification emails through a standard SMTP-compatible email service.

SI-2: Video Conferencing Provider

SI-2.1: The system shall store and display an externally generated join link and schedule for each live class session; the specific provider is TBD (see Appendix C).

3.4 Communications Interfaces

CI-1: All client-server communication shall use HTTPS (TLS).

CI-2: The system shall send email notifications in HTML format with a plain-text fallback.

4. System Features

This section organizes the functional requirements of the E-LMS by system feature, the major services the product must provide.

4.1 Course Creation and Management

4.1.1 Description and Priority

An Instructor may create a new course, organize it into modules/units, and publish learning material for enrolled Students to view. An Instructor may edit or archive a course they own. Priority = high.

4.1.2 Stimulus/Response Sequences

Stimulus: Instructor requests to create a new course.

Response: System prompts the instructor for course title, description, and enrollment capacity.

Stimulus: Instructor requests to publish material to a course module.

Response: System validates the file and adds it to the specified module.

Stimulus: Instructor requests to archive a course.

Response: System hides the course from new enrollment while preserving existing Student access to historical grades and material.

4.1.3 Functional Requirements

Course.Create: The system shall let an Instructor who is logged in create a new course with a title, description, and start/end dates.

Course.Create.Capacity: The system shall allow the Instructor to optionally set a maximum enrollment capacity for the course.

Course.Module.Add: The system shall allow the Instructor to organize course material into one or more named modules or units.

Course.Material.Upload: The system shall allow the Instructor to upload material (documents, slides, video links) to a module.

Course.Material.Type: The system shall reject file uploads that do not match the allowed file types configured for the institution.

Course.Enroll.Student: The system shall allow a Student to enroll in a course that has not reached its enrollment capacity and has not passed its enrollment deadline.

Course.Enroll.Full: If a course has reached its enrollment capacity, the system shall inform the Student and offer a waitlist option, if enabled by the Instructor.

Course.Edit: The system shall allow the Instructor who owns a course to edit its title, description, dates, and material at any time before the course is archived.

Course.Archive: The system shall allow the Instructor to archive a course, which shall prevent new enrollments while preserving read access to existing Students.

4.2 Assignment Submission

4.2.1 Description and Priority

An Instructor may create an assignment with a due date and instructions. An enrolled Student may submit a file before or after the due date; submissions made after the due date are flagged as Late. A Student may resubmit before grading occurs. Priority = high.

4.2.2 Stimulus/Response Sequences

Stimulus: Instructor requests to create an assignment.

Response: System prompts for title, instructions, due date, allowed file types, and maximum marks.

Stimulus: Student requests to submit an assignment.

Response: System accepts the file and records the submission timestamp.

Stimulus: Student requests to submit after the due date has passed.

Response: System accepts the submission but flags it as Late.

4.2.3 Functional Requirements

Assign.Create: The system shall let an Instructor create an assignment with a title, instructions, due date, allowed file types, and maximum marks.

Assign.Visible: The system shall make an assignment visible to enrolled Students only once the Instructor publishes it.

Assign.Submit: The system shall allow an enrolled Student to submit one file per assignment, subject to the allowed file types.

Assign.Submit.Timestamp: The system shall record the exact date and time of each submission.

Assign.Submit.Late: If a submission is made after the due date, the system shall accept it and flag its status as "Late" rather than reject it.

Assign.Submit.Resubmit: The system shall allow a Student to resubmit an assignment, replacing the previous file, at any time before the Instructor has entered a grade for it.

Assign.Submit.LatePenalty: The Instructor may configure a late-penalty rule per assignment; if configured, the system shall apply it automatically when calculating the grade for a Late submission.

Assign.View.Status: The system shall allow a Student to view the submission status (Not Submitted, Submitted, Late, Graded) of each of their assignments.

Assign.Instructor.List: The system shall allow an Instructor to view a list of all submissions for an assignment, including submission time and Late flag.

4.3 Quizzes

4.3.1 Description and Priority

An Instructor may create a timed quiz composed of multiple-choice and true/false questions. An enrolled Student may attempt the quiz within the configured time limit and attempt count; objective questions are graded automatically upon submission. Priority = high.

4.3.2 Stimulus/Response Sequences

Stimulus: Instructor requests to create a quiz.

Response: System prompts for title, time limit, number of allowed attempts, and question list.

Stimulus: Student requests to start a quiz attempt.

Response: System displays the questions and starts a countdown timer.

Stimulus: Quiz time limit expires before the Student submits.

Response: System auto-submits the Student's current answers.

4.3.3 Functional Requirements

Quiz.Create: The system shall let an Instructor create a quiz containing one or more multiple-choice or true/false questions, each with a designated correct answer and point value.

Quiz.Create.Timer: The system shall allow the Instructor to set a time limit for the quiz.

Quiz.Create.Attempts: The system shall allow the Instructor to set the maximum number of attempts a Student may make (default = 1).

Quiz.Create.Randomize: The system shall allow the Instructor to enable randomized question order for each Student attempt.

Quiz.Attempt.Start: The system shall allow an enrolled Student to start a quiz attempt only if the quiz is currently open and the Student has remaining attempts.

Quiz.Attempt.Timer: Once a Student starts an attempt, the system shall display a visible countdown of the remaining time.

Quiz.Attempt.AutoSubmit: If the time limit expires before the Student manually submits, the system shall automatically submit the Student's answers as they stand.

Quiz.Grade.Auto: Upon submission, the system shall automatically grade all multiple choice and true/false answers and display the resulting score to the Student immediately.

Quiz.Attempt.Review: The system shall allow the Student to review their submitted answers and correct answers, if the Instructor has enabled answer review for that quiz.

4.4 Attendance Tracking

4.4.1 Description and Priority

An Instructor may mark attendance for each enrolled Student for a given class session. Students may view their own attendance record. Priority = High.

4.4.2 Stimulus/Response Sequences

Stimulus: Instructor requests to open attendance for a session.

Response: System displays the enrolled Student roster for that session.

Stimulus: Instructor marks a Student's attendance status.

Response: System records the status against that student.

Stimulus: Student requests to view their attendance record.

Response: System displays the Student's attendance history for the course.

4.4.3 Functional Requirements

Attend.Session.Open: The system shall allow an Instructor to open an attendance session for a specific course meeting date.

Attend.Mark: The system shall allow the Instructor to mark each enrolled Student's status for that session as Present, Absent, or Late.

Attend.Mark.Default: The system shall default every Student's status to Absent until explicitly marked otherwise.

Attend.Mark.Bulk: The system shall allow the Instructor to mark all Students Present with a single action, then adjust individual exceptions.

Attend.Edit: The system shall allow the Instructor to edit a previously recorded attendance status up to 7 days after the session date.

Attend.View.Student: The system shall allow a Student to view their own attendance record for each course they are enrolled in.

Attend.Report.Instructor: The system shall allow the Instructor to generate an attendance summary report for a course, showing each Student's attendance percentage.

Attend.Alert.Threshold: If a Student's attendance percentage for a course falls below a configurable threshold, the system shall flag that Student in the Instructor's attendance report.

4.5 Automated Grading and Gradebook

4.5.1 Description and Priority

The system automatically computes and records grades for auto-gradable quiz questions and maintains a per-course gradebook combining quiz and assignment scores. Instructors may enter or override grades for assignments. Priority = High.

4.5.2 Stimulus/Response Sequences

Stimulus: Student submits a quiz.

Response: System calculates and records the score for auto-gradable questions without human intervention.

Stimulus: Instructor enters a grade for a submitted assignment.

Response: System records the grade and updates the Student's overall course grade.

Stimulus: Student requests to view their grades.

Response: System displays all released grades for the course, and the computed overall grade.

4.5.3 Functional Requirements

Grade.Auto.Quiz: The system shall automatically calculate and record a Student's quiz score immediately upon quiz submission, with no Instructor action required.

Grade.Manual.Assign: The system shall allow an Instructor to enter a numeric grade and optional feedback comment for a submitted assignment.

Grade.Weight: The system shall allow the Instructor to define grade-category weights for a course (e.g., Assignments 40%, Quizzes 30%, Attendance 10%, Other 20%).

Grade.Compute.Overall: The system shall compute each Student's overall course grade as the weighted sum of category averages, recalculated whenever a component grade changes.

Grade.Release.Control: The system shall not display a grade to a Student until the Instructor has explicitly released it, even if the grade has already been computed.

Grade.Gradebook.View: The system shall provide the Instructor a gradebook view listing every enrolled Student and their scores across all graded components of the course.

Grade.Export: The system shall allow the Instructor to export the gradebook for a course as a CSV file.

Grade.Student.View: The system shall allow a Student to view only their own released grades, not those of other Students.

4.6 Discussion Forums / Q&A

4.6.1 Description and Priority

Each course has an associated discussion forum where enrolled Students, and the Instructor may post questions, replies, and announcements. Priority = Medium.

4.6.2 Stimulus/Response Sequences

Stimulus: Student posts a new question in a course forum.

Response: System publishes the post and notifies the Instructor assigned to the course.

Stimulus: Instructor replies to a forum post.

Response: System publishes the reply and notifies the original poster.

4.6.3 Functional Requirements

Forum.Post.Create: The system shall allow an enrolled Student or the Instructor to create a new discussion thread within a course's forum.

Forum.Post.Reply: The system shall allow any enrolled course member to reply to an existing thread.

Forum.Post.Edit: The system shall allow the original author of a post to edit or delete it within 15 minutes of posting.

Forum.Moderate: The system shall allow the Instructor to delete or hide any post in a course forum they are assigned to, regardless of author.

Forum.Pin: The system shall allow the Instructor to pin a thread (e.g., an announcement) to the top of the forum.

Forum.Search: The system shall allow a course member to search forum threads within that course by keyword.

4.7 Notifications

4.7.1 Description and Priority

The system sends email notifications to affected users for key course events, including approaching assignment deadlines, released grades, and new forum activity. Priority = Medium.

4.7.2 Stimulus/Response Sequences

Stimulus: An assignment due date is within a configurable reminder window. **Response:** System sends a deadline-reminder email to each Student who has not yet submitted.

Stimulus: An Instructor releases grades for an assignment or quiz.

Response: System sends a grade-posted email to each affected Student.

4.7.3 Functional Requirements

Notify.Deadline: The system shall send an email reminder to each enrolled Student who has not yet submitted an assignment, a configurable number of hours before its due date.

Notify.GradePosted: The system shall send an email to a Student when a grade is released for one of their submissions.

Notify.Forum: The system shall send an email to the original poster of a forum thread when a new reply is added to it.

Notify.Announcement: The system shall send an email to all enrolled Students when the Instructor posts a pinned course announcement.

Notify.Preference: The system shall allow a user to opt out of non-essential notification

categories (forum replies) but not out of deadline and grade notifications.

Notify.LiveClass: The system shall send an email with the join link to enrolled Students at a configurable time before a scheduled live class session.

4.8 Video Conferencing / Live Classes

4.8.1 Description and Priority

An Instructor may schedule a live class session for a course and provide a join link generated by an external video-conferencing provider. Enrolled Students may view scheduled sessions and join at the scheduled time. Priority = medium.

4.8.2 Stimulus/Response Sequences

Stimulus: Instructor requests to schedule a live session.

Response: System prompts for date, time, duration, and join link.

Stimulus: Student views the course's live-session schedule.

Response: System displays upcoming and past sessions with join links active only during the scheduled window.

4.8.3 Functional Requirements

Live.Schedule: The system shall allow an Instructor to schedule a live class session for a course, specifying date, start time, duration, and an external join link.

Live.List: The system shall display a list of upcoming and past live sessions to enrolled Students within the course.

Live.JoinLink.Active: The system shall only display the join link as clickable during the scheduled session window (from 10 minutes before start to the scheduled end time).

Live.Reschedule: The system shall allow the Instructor to reschedule or cancel a live session, and shall notify enrolled Students of the change per Notify.Announcement.

Live.Recording.Link: The system shall allow the Instructor to optionally attach a recording link to a past session for Students who could not attend live.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

PE-1: The system shall support at least 500 concurrent users during peak periods (e.g., assignment deadlines, quiz windows) without degradation of response time beyond the limits below.

PE-2: Web pages shall be fully rendered within 3 seconds on a standard broadband connection.

PE-3: Quiz auto-grading shall display the Student's score within 2 seconds of quiz submission.

PE-4: The system shall support file uploads of up to 50 MB per assignment submission.

5.2 Safety Requirements

No safety requirements have been identified.

5.3 Security Requirements

SE-1: All network transactions involving personally identifiable information or academic records shall be encrypted in transit.

SE-2: Users shall be required to log in to the E-LMS for all operations except viewing publicly listed course catalogs, if enabled.

SE-3: The system shall enforce role-based access control such that a Student can view only their own grades, submissions, and attendance records.

SE-4: Only users designated as Instructor for a specific course may grade submissions or mark attendance for that course.

SE-5: Only users designated as Instructor for a course may create, edit, archive, or delete that course.

SE-6: The system shall lock an account for a configurable cool-down period after five consecutive failed login attempts.

5.4 Software Quality Attributes

Availability — The E-LMS shall be available 99.5% of the time between 6:00 a.m. and midnight local time on institution business days.

Usability — A new Student shall be able to submit an assignment without external help within 5 minutes of first login, guided by the onboarding walkthrough (UD-2).

Reliability — If a Student's session is interrupted while attempting a quiz, the system shall preserve answers already submitted for auto-grading up to the point of interruption.

Maintainability — The system's course, assignment, quiz, and grading modules shall be implemented as logically separated components to allow independent updates.

5.5 Business Rules

BR-1: Only an Instructor may create, edit, archive, or delete a course they own.

BR-2: An Instructor grades submissions and marks attendance only for courses.

BR-3: A grade is visible to a Student only after the Instructor has released it, regardless of when it was computed or entered.

BR-4: A quiz attempt, once auto-submitted or manually submitted, cannot be reopened by the Student.

6. Other Requirements

No database-specific, internationalization, or legal requirements beyond those already stated have been identified for Release 1.0. Internationalization (multi-language UI) is a candidate for a future release.

Appendix A: Glossary

Course — A set of instructional material and activities created by an Instructor; identified by course ID, title, description, start/end dates, and enrollment capacity.

Attendance Record — A Student's attendance status (Present, Absent, or Late) for a single course session.

Grade — A numeric score for a submission or quiz attempt, plus an optional feedback comment and a released flag.

Forum Thread — A discussion topic within a course forum, including author, title, body, pinned flag, and

timestamp.

Live Session — A scheduled live class meeting for a course, including date, time, duration, external join link, and optional recording link.

E-LMS — Learning Management System — the product specified by this document.

TBD — To Be Determined — a placeholder for information not yet available; see Appendix C.

Appendix B: Analysis Models

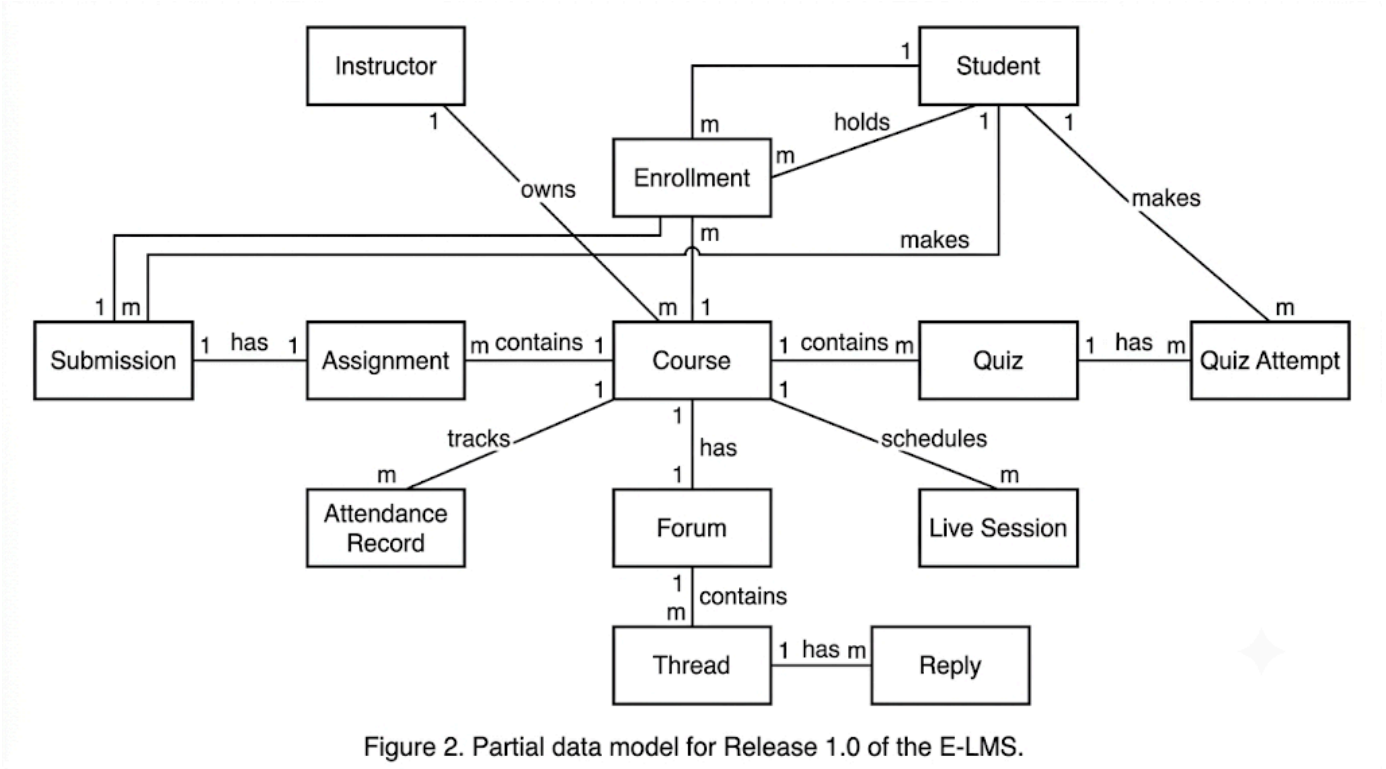


Figure 2. Partial data model for Release 1.0 of the E-LMS.

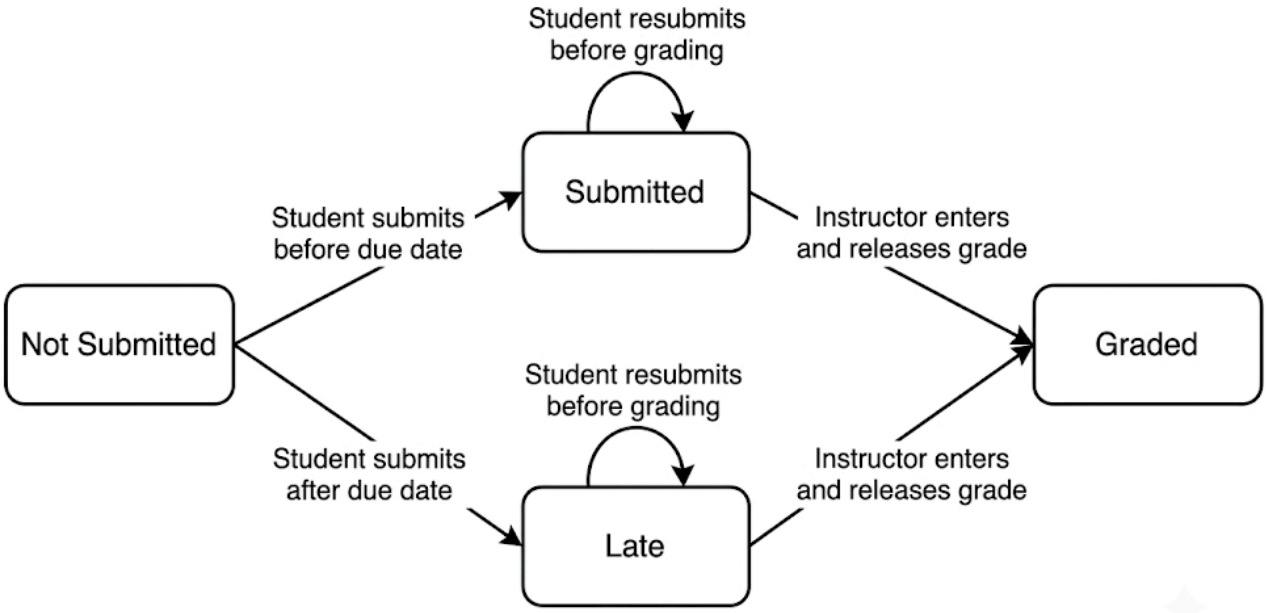


Figure 3. State-transition diagram for Assignment Submission status in the E-LMS.

Appendix C: To Be Determined List

- TBD-1 (Section 2.1 / 3.3): Specific external video-conferencing provider to integrate with (e.g., Zoom, Google Meet) is not yet selected.

- TBD-2 (Section 5.1): Exact peak concurrent-user figure should be revisited once actual institution enrollment numbers are available.

- TBD-3 (Section 4.3): Whether short-answer or file-upload quiz question types will be added in a future release is not yet decided.